



THE
ZSMC CO
ESTD 2026



Appointment of Suresh Pittala

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Effective Date: 30-09-2026

Evidence before prestige. Reproducibility before publicity. Intellectual contribution before title. Scientific honesty before institutional interest.

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Srijal Dutta
Founder
The ZSMC Co.



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Appointment

The ZSMC Co. hereby records the appointment of Dr. Suresh Pittala as a Physics Researcher Advisor within The ZSMC Co. research program.

The appointment follows the review of his Research Program Application. The application was assessed on the basis of the proposed research, scientific merit, feasibility, research integrity, and the applicant's ability to contribute to collaborative research activities.

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Researcher Information

FULL NAME	Dr. Suresh Pittala
Role	Physics Research Advisor
Appointment Status	Final
Email	sureshp@thezsmc.co
Country	India
ORCID	0000-0003-3187-466X
Affiliation	The ZSMC Co.

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Research Area

Dr. Suresh Pittala's research interests and proposed work primarily concern:

- Condensed Matter Physics
- High Entropy Oxides
- Advanced Functional Materials
- Quantum Materials
- Flexible Thermoelectrics
- Multiferroics
- Semiconductor Materials
- Thin films
- Nanotechnology
- Machine Learning-assisted materials
- Design for renewable energy and electronic applications



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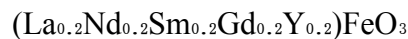
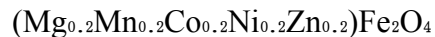
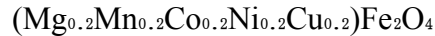


Principal Research Project

The increasing demand for sustainable energy technologies has accelerated research into thermoelectric materials capable of directly converting waste heat into electricity. However, conventional thermoelectric materials such as Bi_2Te_3 and PbTe face challenges including high cost, toxicity, limited thermal stability, and scarcity of constituent elements. High-Entropy Oxides (HEOs), owing to their high configurational entropy, exceptional thermal stability, and enhanced phonon scattering, have recently emerged as a promising class of materials for next-generation thermoelectric applications.

The primary objective of this project is to design, synthesize, and optimize novel high-entropy oxide ceramics with enhanced thermoelectric performance through compositional engineering and machine learning-assisted materials discovery.

The research will focus on the development of single-phase high-entropy spinel and perovskite oxide systems, including compositions such as:



Other compositionally engineered rare-earth-based high-entropy oxides designed for low thermal conductivity and improved electrical transport.

The project aims to systematically investigate the influence of elemental substitution, configurational entropy, lattice distortion, and defect chemistry on phase stability, carrier transport, phonon scattering, and thermoelectric performance.

The proposed materials will be synthesized using scalable ceramic processing techniques such as solid-state reaction, Pechini sol-gel synthesis, and solution combustion methods, followed by controlled calcination and sintering to obtain phase-pure high-density ceramics.

Comprehensive characterization will be performed using:

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Powder X-ray Diffraction (XRD) with Rietveld refinement for crystal structure and phase analysis.

Scanning Electron Microscopy (SEM) with EDS mapping for microstructure and elemental homogeneity.

Transmission Electron Microscopy (TEM) for nanoscale structural analysis.

Raman spectroscopy to investigate lattice vibrations and structural disorder.

X-ray Photoelectron Spectroscopy (XPS) to determine oxidation states and defect chemistry.

Density measurements and grain-size analysis.

Thermoelectric properties will be evaluated over a wide temperature range through measurements of:

Seebeck coefficient (S)

Electrical conductivity (σ)

Thermal conductivity (κ)

Power factor ($S^2\sigma$)

Figure of merit (ZT)

Temperature-dependent electrical transport, Hall-effect measurements, and impedance spectroscopy will also be carried out to understand charge-carrier concentration, mobility, activation energy, and conduction mechanisms.



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Research Integrity

As part of the application, the applicant affirmed that:

- the information submitted in the application is truthful;
- acceptance is not guaranteed;
- research shall be conducted ethically and honestly;
- experimental results, simulations, and theoretical work shall be distinguished accurately; and
- The ZSMC Co. may decline or discontinue participation where these principles are not followed.

These commitments form part of the basis of the provisional appointment.

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Research Independence and Collaboration

The researcher may continue to pursue independent academic or research activities outside The ZSMC Co., subject to applicable intellectual-property, confidentiality, authorship, and conflict-of-interest requirements.

Participation in The ZSMC Co. does not by itself transfer ownership of independently developed work to the organization.

Where research is conducted collaboratively under The ZSMC Co., ownership, authorship, publication, data use, and intellectual-property arrangements should be established appropriately for the relevant project before material collaborative work is undertaken.

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Researcher Recognition

During the provisional period, Dr. Suresh Pittala may add an affiliation to any related work as follows if he wishes:

Dr. Suresh Pittala — Physics Research Advisor, The ZSMC Co.

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Effective Date and Duration

Effective Date: 30 September, 2026

Duration: Currently indefinite

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Record of Approval

This record constitutes the formal organizational record of the appointment of Dr. Suresh Pittala to the position of Researcher at The ZSMC Co.

Appointing Authority:

Srijal Dutta

Founder & Lead Researcher

The ZSMC Co.

Appointment Decision: ACCEPTED

Date of Decision: 30 September, 2026

Researcher Acknowledgement: Acknowledged

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